

The Rotating Hoop and a Singular Overdamped Limit

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Visualization: Rotating-hoop variables

symbol	meaning
$m > 0$	bead mass
$a > 0$	hoop radius
$g > 0$	gravitational acceleration
$\omega \geq 0$	constant angular speed of the hoop
$b > 0$	viscous-damping coefficient
ϕ	angle from the downward vertical
$a \sin \phi$	signed radial coordinate from the rotation axis

The angular coordinate satisfies

$$-\pi < \phi \leq \pi, \quad -\pi \equiv \pi.$$

Question: What is the rotating-hoop equation?

A bead moves on a vertical circular hoop rotating about its vertical diameter with angular speed ω . Resolve gravity, centrifugal force, and viscous drag in the direction of increasing ϕ .

Solution. The tangential acceleration is $a\ddot{\phi}$. The tangential force components are

force	tangential component
gravity	$-mg \sin \phi$
centrifugal	$m\omega^2(a \sin \phi) \cos \phi$
viscous drag	$-ba\dot{\phi}$.

Newton's law gives

$$ma\ddot{\phi} = -ba\dot{\phi} - mg \sin \phi + m\omega^2 a \sin \phi \cos \phi.$$

Question: How is the hoop equation nondimensionalized?

For

$$ma\ddot{\phi} = -ba\dot{\phi} - mg \sin \phi + ma\omega^2 \sin \phi \cos \phi,$$

choose the slow time scale

$$T = \frac{ba}{mg}, \quad \tau = \frac{t}{T},$$

and derive the dimensionless equation.

Solution. With primes denoting $d/d\tau$,

$$\dot{\phi} = \frac{1}{T}\phi', \quad \ddot{\phi} = \frac{1}{T^2}\phi''.$$

After division by mg ,

$$\frac{a}{gT^2}\phi'' = -\frac{ba}{mgT}\phi' - \sin \phi + \frac{a\omega^2}{g} \sin \phi \cos \phi.$$

The choice of T makes $ba/(mgT) = 1$. Define

$$\varepsilon = \frac{m^2g}{b^2a}, \quad \gamma = \frac{a\omega^2}{g}.$$

Then

$$\varepsilon\phi'' = -\phi' - \sin \phi + \gamma \sin \phi \cos \phi.$$

The strong-damping regime is the limit $\varepsilon \downarrow 0$ with the other dimensionless parameters fixed.

Definition: Overdamped limit

For

$$\varepsilon\phi'' = -\phi' - \sin \phi + \gamma \sin \phi \cos \phi,$$

the limit $\varepsilon \rightarrow 0$ is the **overdamped limit** because damping dominates inertia.

Definition: Singular limit

A parameter limit that lowers the order of the differential equation is called a **singular limit**. The reduced equation then has fewer independent initial conditions than the original equation.

Definition: Initial layer

The **initial layer** is the short initial interval during which the inertial term discarded by the overdamped reduction remains important.

Notation: Fast time

For $\varepsilon > 0$, the **fast time** is

$$s = \frac{\tau}{\varepsilon}$$

so a fixed interval $0 \leq s \leq S$ corresponds to $0 \leq \tau \leq \varepsilon S$.

Question: What is the reduced overdamped flow?

For

$$\varepsilon \phi'' = -\phi' - \sin \phi + \gamma \sin \phi \cos \phi,$$

set $\varepsilon = 0$, find all equilibria for $\gamma \geq 0$, and classify them.

Solution. The reduced equation is

$$\phi' = f(\phi) = \sin \phi (\gamma \cos \phi - 1).$$

The bottom $\phi = 0$ and top $\phi = \pi$ are always equilibria. Since

$$f'(\phi) = \cos \phi (\gamma \cos \phi - 1) - \gamma \sin^2 \phi,$$

we have

$$f'(0) = \gamma - 1, \quad f'(\pi) = \gamma + 1 > 0.$$

Thus the top is always unstable, while the bottom is asymptotically stable for $\gamma < 1$ and unstable for $\gamma > 1$.

If $\gamma > 1$, there are two additional equilibria

$$\phi_{\pm}^* = \pm \arccos(1/\gamma).$$

At either one,

$$f'(\phi_{\pm}^*) = -\gamma \sin^2 \phi_{\pm}^* < 0,$$

so both are asymptotically stable.

Near $(\phi, \gamma) = (0, 1)$,

$$f(\phi) = (\gamma - 1)\phi - \frac{1}{2}\phi^3 + O(|\gamma - 1||\phi|^3 + |\phi|^5) \quad ((\phi, \gamma) \rightarrow (0, 1)).$$

Hence $\gamma = 1$ is a supercritical pitchfork. At $\gamma = 1$,

$$\phi' = -\frac{1}{2}\phi^3 + O(\phi^5) \quad (\phi \rightarrow 0).$$

The flow points toward 0 on both sides, so the bottom is nonhyperbolic and asymptotically stable.

Visualization: Reduced equilibria versus rotation rate

γ	bottom and symmetry-broken states	top
$0 \leq \gamma < 1$	$\phi = 0$ asymptotically stable	$\phi = \pi$ unstable
$\gamma = 1$	$\phi = 0$ nonhyperbolic asymptotically stable	$\phi = \pi$ unstable
$\gamma > 1$	$\phi = 0$ unstable, $\phi = \pm \arccos(1/\gamma)$ asymptotically stable	$\phi = \pi$ unstable

The two stable nonzero branches satisfy

$$\cos \phi_{\pm}^* = \frac{1}{\gamma}, \quad \phi_{\pm}^* \rightarrow \pm \frac{\pi}{2} \quad (\gamma \rightarrow \infty).$$

Question: Where does the missing initial condition go?

Consider

$$\varepsilon y'' + 2y' + y = 0, \quad 0 < \varepsilon < 1, \quad y(0) = y_0, \quad y'(0) = v_0,$$

where primes denote differentiation with respect to τ and y_0, v_0 are fixed. Find the two time scales as $\varepsilon \downarrow 0$ and explain how the solution satisfies both initial conditions.

Solution. The characteristic roots are

$$\lambda_{\pm} = \frac{-1 \pm \sqrt{1 - \varepsilon}}{\varepsilon}.$$

Using

$$\sqrt{1 - \varepsilon} = 1 - \frac{\varepsilon}{2} - \frac{\varepsilon^2}{8} + O(\varepsilon^3),$$

as $\varepsilon \downarrow 0$, we obtain

$$\lambda_+ = -\frac{1}{2} - \frac{\varepsilon}{8} + O(\varepsilon^2), \quad \lambda_- = -\frac{2}{\varepsilon} + \frac{1}{2} + O(\varepsilon).$$

Thus

$$y(\tau) = Ae^{\lambda_+ \tau} + Be^{\lambda_- \tau}.$$

The initial conditions give

$$A + B = y_0, \quad \lambda_+ A + \lambda_- B = v_0,$$

so

$$B = \frac{\lambda_+ y_0 - v_0}{\lambda_+ - \lambda_-}, \quad A = y_0 - B.$$

The corresponding characteristic decay times are

$$\frac{1}{|\lambda_+|} = 2 + O(\varepsilon), \quad \frac{1}{|\lambda_-|} = \frac{\varepsilon}{2} + O(\varepsilon^2) \quad (\varepsilon \downarrow 0).$$

The reduced equation $2y' + y = 0$ forces $y'(0) = -y_0/2$. The fast B term allows the full solution to have any prescribed slope v_0 , then decays on a time interval comparable to ε . The A term evolves on a time interval comparable to 1.

Definition: Slow curve

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. For the system

$$\phi' = \Omega, \quad \varepsilon \Omega' = f(\phi) - \Omega, \quad \varepsilon > 0,$$

the **slow curve of the singular problem** is the algebraic set $\Omega = f(\phi)$ obtained by setting $\varepsilon = 0$.

Proposition: Leading fast-layer relaxation

Fix $\gamma \geq 0$. Write $\Omega = \phi'$ and $f(\phi) = \sin \phi(\gamma \cos \phi - 1)$. Fix $S > 0$ and initial data (ϕ_0, Ω_0) independent of ε . For each $\varepsilon > 0$, let (ϕ, Ω) solve

$$\phi' = \Omega, \quad \varepsilon \Omega' = f(\phi) - \Omega, \quad (\phi(0), \Omega(0)) = (\phi_0, \Omega_0).$$

On the fast time $s = \tau/\varepsilon$, write $\phi_\varepsilon(s) = \phi(\varepsilon s)$ and $\Omega_\varepsilon(s) = \Omega(\varepsilon s)$. Uniformly for $0 \leq s \leq S$ as $\varepsilon \downarrow 0$,

$$\begin{aligned} \phi_\varepsilon(s) &= \phi_0 + O(\varepsilon), \\ \Omega_\varepsilon(s) &= f(\phi_0) + (\Omega_0 - f(\phi_0))e^{-s} + O(\varepsilon). \end{aligned}$$

Thus trajectories relax toward the slow curve $\Omega = f(\phi)$ on the initial layer.

Proof. In fast time,

$$\frac{d\phi_\varepsilon}{ds} = \varepsilon \Omega_\varepsilon, \quad \frac{d\Omega_\varepsilon}{ds} = f(\phi_\varepsilon) - \Omega_\varepsilon.$$

Multiplying the second equation by e^s and integrating gives

$$\Omega_\varepsilon(s) = e^{-s}\Omega_0 + \int_0^s e^{-(s-v)} f(\phi_\varepsilon(v)) dv.$$

Since $|f(\phi)| \leq 1 + \gamma$,

$$|\Omega_\varepsilon(s)| \leq \max\{|\Omega_0|, 1 + \gamma\}.$$

Integrating the first equation gives

$$|\phi_\varepsilon(s) - \phi_0| \leq \varepsilon S \max\{|\Omega_0|, 1 + \gamma\} \quad (0 \leq s \leq S).$$

Since $|f'(\phi)| \leq 1 + \gamma$, the mean-value theorem now gives $f(\phi_\varepsilon(s)) = f(\phi_0) + O(\varepsilon)$ uniformly on $[0, S]$. Substitute this into the integral and use $\int_0^s e^{-(s-v)} dv = 1 - e^{-s}$ to obtain the formula for Ω_ε . $\square$

Question: What is the fast-layer duration in physical time?

With $s = \tau/\varepsilon$ and $t = T\tau$, what physical time corresponds to $s = 1$?

Solution. Since $t = T\tau$,

$$T\varepsilon = \frac{ba}{mg} \frac{m^2g}{b^2a} = \frac{m}{b}, \quad \frac{m/b}{ba/(mg)} = \frac{m^2g}{b^2a} = \varepsilon.$$

Thus m/b is the inertial damping time, smaller than the slow time $T = ba/(mg)$ by the factor ε .